

OOPs assignment 7

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Q1 Class polygon contains data member width and height and public method set_value() to assign values to width and height. Class Rectangle and Triangle are inherited from polygon class. Both the classes contain public method calculate_area() to calculate the area of Rectangle and Triangle. Use base class pointer to access the derived class object and show the area calculated.

ain.cpp

```
1  #include <iostream>
2  using namespace std;
3
4  class Polygon {
5  protected:
6      int width, height;
7  public:
8      void set_value(int w, int h) {
9          width = w;
10         height = h;
11     }
12     virtual int area() = 0;
13 };
14
15 class Rectangle : public Polygon {
16 public:
17     int area() {
18         return width * height;
19     }
20 };
21
22 class Triangle : public Polygon {
23 public:
24     int area() {
25         return (width * height) / 2;
26     }
27 };
28
29 int main() {
30     Polygon *p;
```

```

input
Rectangle area = 50
Triangle area = 25

...Program finished with exit code 0
Press ENTER to exit console.

```

Q2 Write a program to create a class shape with functions *area* and *display* to find area and display the name of the shape and other essential component of the class. Create derived classes circle, rectangle and triangle each having overridden functions *area* and *display*. Write a program to find and display the area of circle, rectangle and triangle

main.cpp

```
1  #include <iostream>
2  using namespace std;
3
4  class Shape {
5  public:
6      virtual void area() = 0;
7      virtual void display() = 0;
8  };
9
10 class Circle : public Shape {
11     float r;
12 public:
13     Circle(float r) { this->r = r; }
14     void area() { cout << "Area = " << 3.14 * r * r << endl; }
15     void display() { cout << "Circle\n"; }
16 };
17
18 class Rectangle : public Shape {
19     int l, b;
20 public:
21     Rectangle(int l, int b) { this->l = l; this->b = b; }
22     void area() { cout << "Area = " << l * b << endl; }
23     void display() { cout << "Rectangle\n"; }
24 };
25
26 class Triangle : public Shape {
27     int b, h;
28 public:
29     Triangle(int b, int h) { this->b = b; this->h = h; }
30     void area() { cout << "Area = " << 0.5 * b * h << endl; }
31     void display() { cout << "Triangle\n"; }
32 };
33
34 int main() {
35     Shape *s;
36
37     Circle c(5);
38     Rectangle r(4,6);
39     Triangle t(3,7);
40
41     s = &c; s->display(); s->area();
42     s = &r; s->display(); s->area();
43     s = &t; s->display(); s->area();
44 }
```

```
input
Circle
Area = 78.5
Rectangle
Area = 24
Triangle
Area = 10.5

...Program finished with exit code 0
Press ENTER to exit console.
```

Q3 Write a C++ program to compute area of right angle triangle, equilateral triangle, isosceles triangle using function overloading concept.

```
main.cpp
1  #include <iostream>
2  #include <cmath>
3  using namespace std;
4
5  class Triangle {
6  public:
7      double area(double b, double h) { return 0.5 * b * h; } // right
8
9      double area(double a) { return (sqrt(3)/4) * a * a; } // equilateral
10
11     double area(double a, double b, double c) { // isosceles (Heron)
12         double s = (a+b+c)/2;
13         return sqrt(s*(s-a)*(s-b)*(s-c));
14     }
15 };
16
17 int main() {
18     Triangle t;
19     cout << t.area(10,5) << endl;
20     cout << t.area(6) << endl;
21     cout << t.area(5,5,6) << endl;
22 }
```

A screenshot of a console window with a black background and white text. The window title is 'input'. The output shows three lines of input: '25', '15.5885', and '12'. Below these, a green message states '...Program finished with exit code 0' followed by 'Press ENTER to exit console.' in white.

```
input
25
15.5885
12

...Program finished with exit code 0
Press ENTER to exit console.
```

Q4 Write a program with Student as abstract class and create derive classes Engineering, Medicine and Science from base class Student. Create the objects of the derived classes and process them and access them using array of pointer of type base class Student.

main.cpp

```
1  #include <iostream>
2  using namespace std;
3
4  class Student {
5  public:
6      virtual void show() = 0;
7  };
8
9  class Engineering : public Student {
10 public:
11     void show() { cout << "Engineering\n"; }
12 };
13
14 class Medicine : public Student {
15 public:
16     void show() { cout << "Medicine\n"; }
17 };
18
19 class Science : public Student {
20 public:
21     void show() { cout << "Science\n"; }
22 };
23
24 int main() {
25     Student *s[3];
26
27     Engineering e;
28     Medicine m;
29     Science sc;
30 }
```

```

11     void show() { cout << "Engineering\n"; }
12 };
13
14 class Medicine : public Student {
15 public:
16     void show() { cout << "Medicine\n"; }
17 };
18
19 class Science : public Student {
20 public:
21     void show() { cout << "Science\n"; }
22 };
23
24 int main() {
25     Student *s[3];
26
27     Engineering e;
28     Medicine m;
29     Science sc;
30
31     s[0] = &e;
32     s[1] = &m;
33     s[2] = &sc;
34
35     for(int i=0;i<3;i++)
36         s[i]->show();
37 }

```

```

input
Engineering
Medicine
Science

...Program finished with exit code 0
Press ENTER to exit console.

```

Q5 Create a class Time with three private variables int h,m,s; Create a function to overload '+' operator to add two time variables.

```

int main(){
    Time t1(5,15,34),t2(9,53,58),t3;
    t3 = t1 + t2; t3.show();
}

```


in.cpp

```
1  #include <iostream>
2  using namespace std;
3
4  class Time {
5      int h,m,s;
6  public:
7      Time(int h=0,int m=0,int s=0) {
8          this->h=h; this->m=m; this->s=s;
9      }
10
11     Time operator+(Time t) {
12         Time temp;
13         temp.s = s + t.s;
14         temp.m = m + t.m + temp.s/60;
15         temp.s %= 60;
16         temp.h = h + t.h + temp.m/60;
17         temp.m %= 60;
18         return temp;
19     }
20
21     void show() {
22         cout << h << ":" << m << ":" << s << endl;
23     }
24 };
25
26 int main() {
27     Time t1(5,15,34), t2(9,53,58), t3;
28     t3 = t1 + t2;
29     t3.show();
30 }
```



input

15:9:32

...Program finished with exit code 0
Press ENTER to exit console.

Q6 Define a class STRING and overload == to compare two strings and + operator for concatenation of two strings.

```
main.cpp
1  #include <iostream>
2  #include <string>
3  using namespace std;
4
5  class STRING {
6      string s;
7  public:
8      STRING(string s="") { this->s=s; }
9
10     bool operator==(STRING obj) {
11         return s == obj.s;
12     }
13
14     STRING operator+(STRING obj) {
15         return STRING(s + obj.s);
16     }
17
18     void display() { cout << s << endl; }
19 };
20
21 int main() {
22     STRING a("Hello "), b("World");
23
24     STRING c = a + b;
25     c.display();
26
27     if(a == b)
28         cout << "Equal\n";
29     else
30         cout << "Not Equal\n";}
```

```
input
Hello World
Not Equal

...Program finished with exit code 0
Press ENTER to exit console.
```

Q7 Write a program in C++ to create a class matrix and overload * operator using friend function to multiply two matrices.

```
ain.cpp
1  #include <iostream>
2  using namespace std;
3
4  class Matrix {
5      int a[2][2];
6  public:
7      void input() {
8          for(int i=0;i<2;i++)
9              for(int j=0;j<2;j++)
10                 cin >> a[i][j];
11      }
12
13     void display() {
14         for(int i=0;i<2;i++){
15             for(int j=0;j<2;j++)
16                 cout << a[i][j] << " ";
17             cout << endl;
18         }
19     }
20
21     friend Matrix operator*(Matrix, Matrix);
22 };
23
24 Matrix operator*(Matrix m1, Matrix m2) {
25     Matrix res;
26     for(int i=0;i<2;i++)
27         for(int j=0;j<2;j++) {
28             res.a[i][j]=0;
```

```

18     }
19 }
20
21 friend Matrix operator*(Matrix, Matrix);
22 };
23
24 Matrix operator*(Matrix m1, Matrix m2) {
25     Matrix res;
26     for(int i=0;i<2;i++)
27     for(int j=0;j<2;j++) {
28         res.a[i][j]=0;
29         for(int k=0;k<2;k++)
30             res.a[i][j]+=m1.a[i][k]*m2.a[k][j];
31     }
32     return res;
33 }
34
35 int main() {
36     Matrix m1,m2,m3;
37     m1.input();
38     m2.input();
39     m3 = m1 * m2;
40     m3.display();
41 }

```

input

```

1
2
3
4
5
6
7
8
19 22
43 50

...Program finished with exit code 0
Press ENTER to exit console.

```

